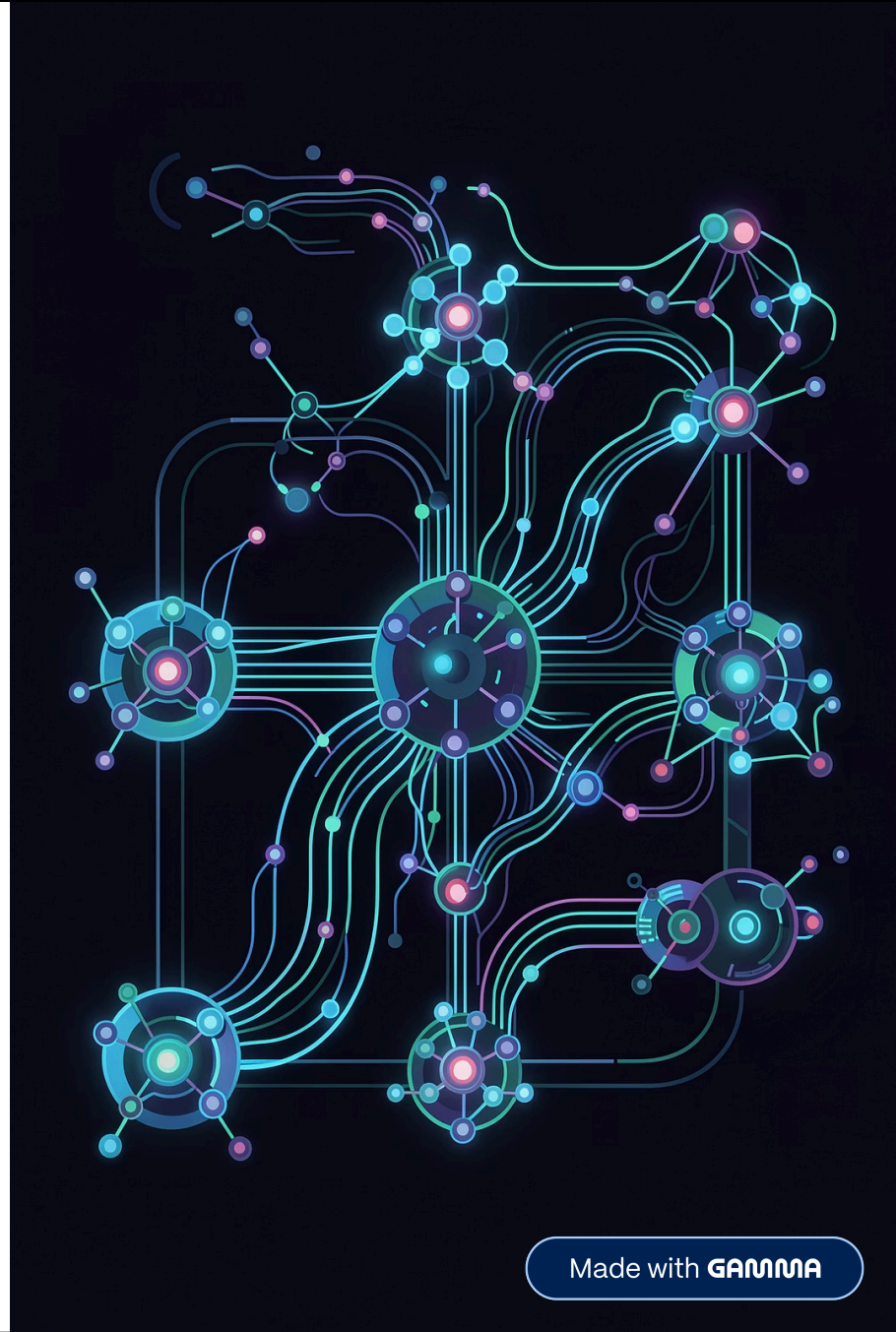
DS Mentor Pro

Agentic Data Science Tutor

A stage-aware mentoring assistant that guides learners through the full machine learning pipeline — combining retrieval grounding, intent routing, structured teaching responses, and optional code generation.

Group 2



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Project Goal

DS Mentor Pro is a **stage-aware agentic tutor** for the 7-stage ML pipeline. It routes learner queries, retrieves grounded context, and returns structured teaching responses — with optional code generation and anti-pattern detection.

01

Problem Understanding

02

Data Collection

03

Data Preparation

04

Modeling

05

Evaluation

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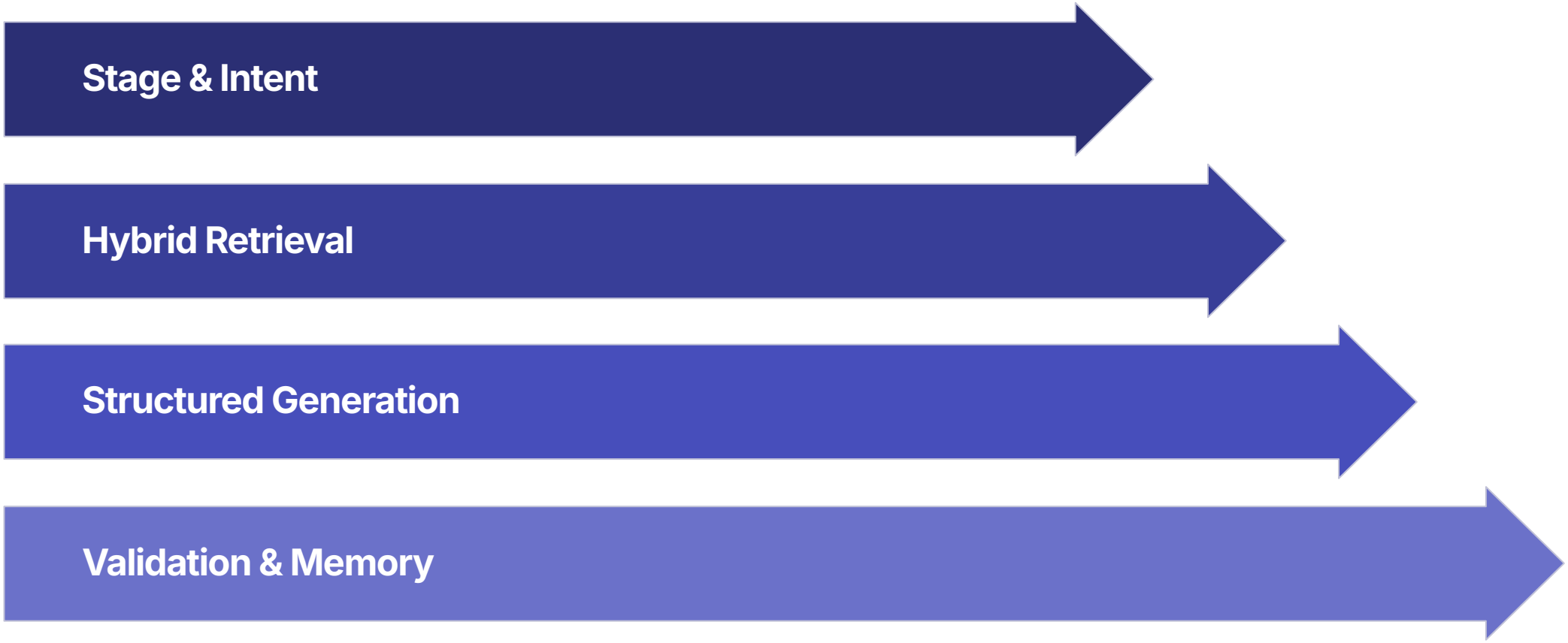
System Architecture



The agentic loop is stage-conditioned at every step — retrieval, generation, and validation all adapt to the learner's current position in the ML pipeline.

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Dataset Construction

Ingestion

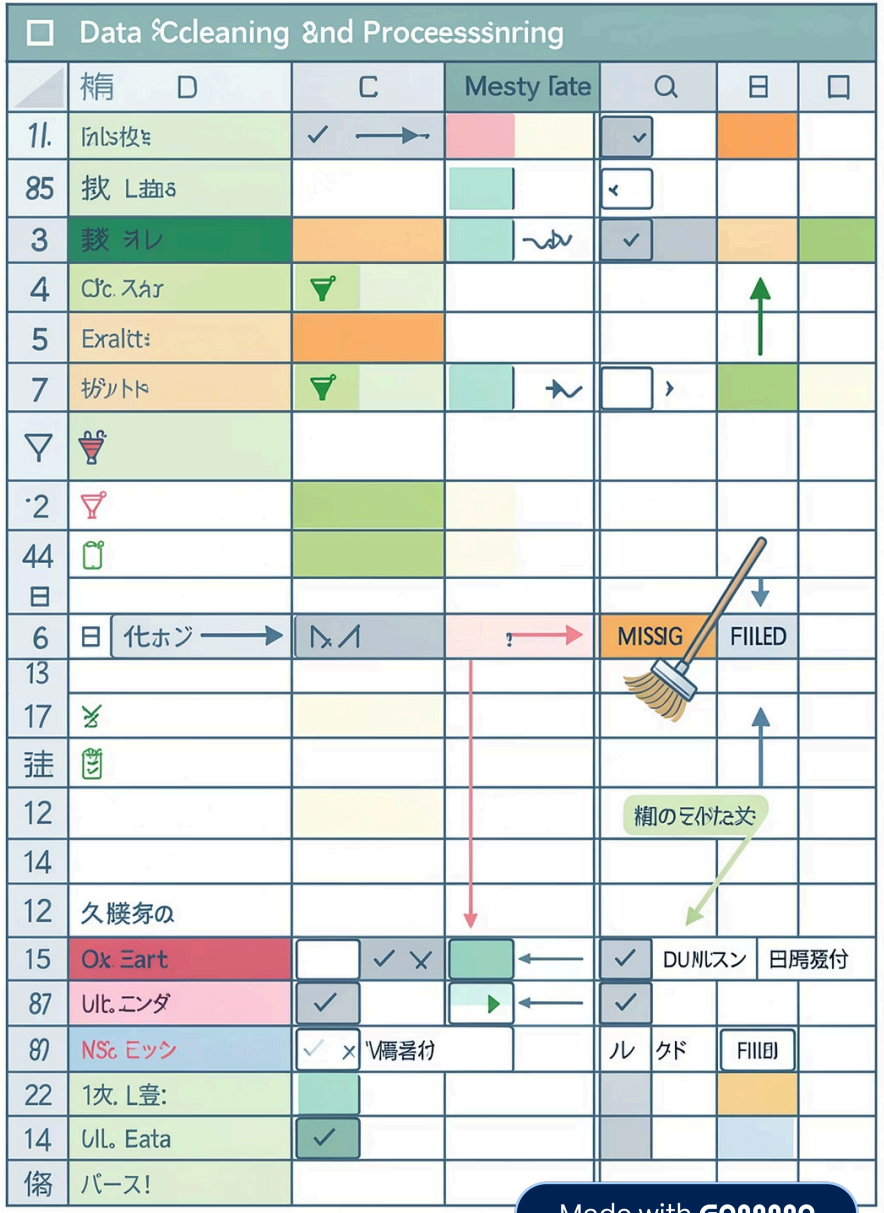
Kaggle notebook ingestion (when enabled) merged with curated fallback sources.

Cleaning

Deduplication, normalization, and stratified train/val/test splits.

Scale

4,856 rows in our expanded dataset, serialized to runtime_dataset.csv.



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Stage Classification: What We Ship

Primary Classifier (Default)

TF-IDF + calibrated LinearSVM — fast, reproducible, strong baseline for strict pipeline runs. Probability calibration enables confident routing decisions.

> ⓘ Selected for reproducibility and auditability in academic evaluation settings.

Optional Transformer Path

A transformer-based classifier exists in the codebase but is **not the default** in strict reproducible runs — reserved for ablation and future work.

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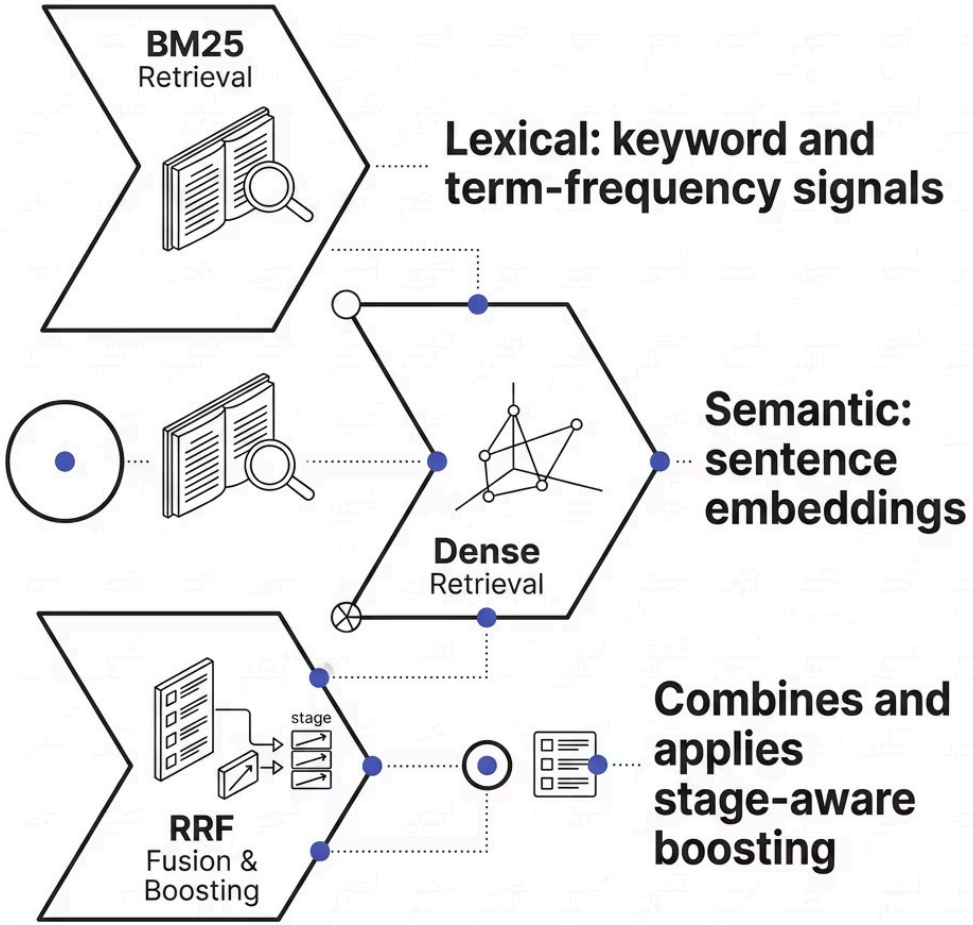
Hybrid Retrieval: BM25 + RRF + Stage Boosting

BM25 Intuition

BM25 scores term relevance with saturation:

$$\text{score}(q, d) = \sum_i \text{IDF}(q_i) \cdot \frac{f(q_i, d) \cdot (k_1 + 1)}{f(q_i, d) + k_1(1 - b + b \cdot |d|/L)}$$

Reciprocal Rank Fusion merges BM25 and dense rankings without tuning. Stage-aware boosting promotes documents matching the learner's current pipeline stage.



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Code Generation

Model

Salesforce CodeT5-small fine-tuned for explanation → code, conditioned on stage context.

Fallback Chain

- Primary: fine-tuned CodeT5-small
- Fallback: base CodeT5
- Last resort: template generator

Syntax Validation

All outputs validated via **AST parsing** before delivery — malformed code is caught and regenerated.

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Retrieval Evaluation Metrics

Metric	Value
Precision@1	0.5000
Precision@3	0.1667
Precision@5	0.1000
Recall@1	0.5000
Recall@3	0.5000
Recall@5	0.5000
MRR	0.5000
nDCG@5	0.5000

Source: outputs/eval_suite_report.md — produced by `python -m evaluation.run_suite` on `evaluation/datasets/small_eval.jsonl`.

- Recall@5 = 0.50 indicates relevant documents often appear within top-5, but not always ranked at the very top.
- Lower Precision@3/@5 suggests ranking quality at deeper positions can be improved (more noise in ranks 3–5).

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Evaluation: Code Generation Results

Results from `codet5_split_eval.json` (val/test) and `codet5_benchmark_eval.json` (10-query regression). CodeBLEU-style is a lightweight approximation, not the official metric.

Split	Syntax	Token Overlap	Line Match	BLEU-1	CodeBLEU-style	Coverage
Val (726)	0.9904	0.2925	0.2211	0.2965	0.5915	726/726
Test (734)	0.9864	0.3219	0.2509	0.3242	0.6075	734/734
Benchmark (10)	1.0000	0.0308	0.0000	0.0290	0.6114	10/10

Benchmark overlap is low — references are short snippets; lexical metrics understate functional correctness.

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Limitations & Future Work

Current Limitations

- Hosted notebook dependency and HF stack version sensitivity
- Retrieval ranking quality degrades at higher k
- Benchmark references are short snippets — lexical metrics understate usefulness
- Evaluation suite sample sizes differ by script

Future Directions

- Cross-encoder reranking for improved retrieval precision
- Larger labeled evaluation sets and unified harness
- Stronger classifier experiments (transformer path)
- Controlled LLM augmentation with grounding checks

DS Mentor Pro demonstrates that stage-aware agentic tutoring — combining hybrid retrieval, calibrated classification, and validated code generation — can deliver structured, reproducible guidance across the full ML pipeline, advancing scalable DS education.

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